Intro to Machine Learning
Practice Problems 4: QDA and Naive Bayes
Practice problems, not turned in — Solutions

Directions: These practice problems go with Week 3 (quadratic discriminant analysis and naive Bayes). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording. Practice Problems 3 covers logistic regression and LDA.

Notation follows ISL 4.4. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. QDA assumes $X \mid Y = k \sim N(\mu_k, \Sigma_k)$; naive Bayes assumes $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$.

1. **QDA is quadratic.** (ISL 4.8, Exercise 3; ISL 4.4.3, ESL 4.3.) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$, with a different variance in each class. Find the discriminant $\delta_k(x)$ the Bayes classifier maximizes, and show that the decision boundary between two classes is not linear in x unless $\sigma_1 = \sigma_2$.

   Solution. As for LDA, the Bayes classifier maximizes $\log \pi_k + \log f_k(x)$. Now

   $$\log \pi_k + \log f_k(x) = \log \pi_k - \tfrac{1}{2}\log(2\pi) - \log \sigma_k - \frac{(x - \mu_k)^2}{2\sigma_k^2}.$$

   Only $-\frac{1}{2}\log(2\pi)$ is the same for every class, so

   $$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log \sigma_k + \log \pi_k.$$

   The boundary between classes 1 and 2 is $\delta_1(x) = \delta_2(x)$, and the x^2 coefficient of $\delta_1(x) - \delta_2(x)$ is $\frac{1}{2}(1/\sigma_2^2 - 1/\sigma_1^2)$. It is zero only if $\sigma_1 = \sigma_2$. Otherwise the boundary is the solution of a quadratic, so it can be two points, and the classification regions need not be intervals.

2. **QDA by hand: same means, different spreads.** (ISL 4.4.3.) Two equally likely classes have $X \mid Y = 1 \sim N(0, 1)$ and $X \mid Y = 2 \sim N(0, 4)$.

   (a) Find the Bayes decision rule. Where is each class predicted?

   (b) Now let $\pi_1 = 0.8$. How does the rule change?

   Solution.

(a) From Problem 1 with $\mu_1 = \mu_2 = 0$, $\sigma_1 = 1$, $\sigma_2 = 2$ and equal priors (which cancel), $\delta_1(x) = -x^2/2$ and $\delta_2(x) = -x^2/8 - \log 2$. Predict class 1 when

$$-\frac{x^2}{2} > -\frac{x^2}{8} - \log 2 \iff \frac{3x^2}{8} < \log 2 \iff |x| < \sqrt{\tfrac{8 \log 2}{3}} = 1.36.$$

Class 1 is predicted on $(-1.36, 1.36)$ and class 2 on both sides of it. The boundary is two points, and class 2's region is two separate pieces.

(b) The prior term $\log(\pi_1/\pi_2) = \log 4$ now favours class 1: predict class 1 when $3x^2/8 < \log 2 + \log 4 = \log 8$, that is $|x| < \sqrt{8 \log 8/3} = 2.35$. Class 1's interval widens.

3. **LDA or QDA?** (ISL 4.8, Exercise 5; ISL 4.4.3, 4.5.) Answer each and explain.

(a) If the Bayes decision boundary is linear, do we expect LDA or QDA to do better on the training set? On the test set?

(b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to do better on the training set? On the test set?

(c) In general, as n increases, do we expect the test accuracy of QDA relative to LDA to improve, decline or stay the same?

(d) True or false: even if the Bayes boundary is linear, QDA will probably achieve a lower test error rate than LDA, because QDA is flexible enough to model a linear boundary.

(e) With $p = 10$ predictors and $K = 3$ classes, how many covariance parameters do LDA, QDA and Gaussian naive Bayes each estimate? How many observations does each class need for QDA's $\hat{\Sigma}_k$ to be invertible?

Solution.

(a) QDA on the training set: it is more flexible and can always match LDA's boundary, so it fits the training data at least as well. LDA on the test set: a linear boundary is right, so QDA's extra flexibility buys no reduction in bias, only extra variance.

(b) QDA on the training set (for the same reason as in (a)). On the test set, usually QDA, if n is large enough to estimate the Σ_k well: LDA's linear boundary is biased, and that bias doesn't shrink with n. With small n and large p, LDA can still win.

(c) Improve. QDA's disadvantage is variance, which shrinks as n grows; any advantage it has is lower bias, which doesn't.

(d) False. If the Bayes boundary is linear, LDA's model is correct, so QDA's extra parameters add variance and no reduction in bias. QDA *can* represent the linear boundary, but its estimate will not be exactly linear, and it pays for the extra parameters in test error.

(e) Each Σ has $p(p+1)/2 = 55$ distinct entries. LDA: 55 (one shared matrix). QDA: $3 \times 55 = 165$. Gaussian naive Bayes: $Kp = 30$ variances (only the diagonal of each Σ_k). All three also estimate $Kp = 30$ means. A sample covariance computed from n_k points has rank at most $n_k - 1$, so QDA needs $n_k \geq p + 1 = 11$ in *every* class.

4. **Naive Bayes with categorical predictors, by hand.** (ISL 4.4.4.) Twelve students took a quiz. For each we record whether they attended the lecture, how many of the practice problems they did (none, some or all), and whether they slept at least 7 hours:

Student	Result	Attended	Practice problems	Slept 7h+
1	pass	yes	all	yes
2	pass	yes	all	no
3	pass	yes	some	yes
4	pass	yes	some	yes
5	pass	no	all	yes
6	pass	yes	none	yes
7	pass	no	some	no
8	pass	yes	some	no
9	fail	no	none	no
10	fail	yes	none	no
11	fail	no	some	yes
12	fail	no	none	no

(a) Estimate the priors, and for each class the proportion of students in each category of each predictor.

(b) Student A attended, did some of the practice problems and slept less than 7 hours. Find the naive Bayes estimate of $\Pr(\text{pass} \mid x)$. (The fractions work out exactly.)

(c) Student B did not attend, did none of the problems and slept less than 7 hours. Find $\Pr(\text{pass} \mid x)$.

(d) Student C did not attend, did all of the problems and slept less than 7 hours. What does naive Bayes give, and why is it a problem? Is C's combination of predictor values in the training data?

(e) Redo (d) with Laplace smoothing: add 1 to every count, so a predictor with m categories has $\hat{f}_{kj}(v) = (\text{count} + 1)/(n_k + m)$. Leave the priors unsmoothed.

(f) Without the naive Bayes assumption, how many probabilities would you have to estimate per class to describe the joint distribution of the three predictors? How many does naive Bayes estimate? Why does that matter with 12 students?

Solution.

(a) $\hat{\pi}_{\text{pass}} = 8/12 = 2/3$ and $\hat{\pi}_{\text{fail}} = 4/12 = 1/3$. Within each class:

Class	Attended	None	Some	All	Slept 7h+
pass ($n_k = 8$)	6/8	1/8	4/8	3/8	5/8
fail ($n_k = 4$)	1/4	3/4	1/4	0/4	1/4

"Attended" and "Slept 7h+" are the proportions answering yes; the "no" proportions are one minus these. "None", "Some" and "All" are the practice-problem categories.

(b) Multiply the prior by one factor per predictor:

$$\text{pass: } \tfrac{2}{3} \cdot \tfrac{6}{8} \cdot \tfrac{4}{8} \cdot \tfrac{3}{8} = \tfrac{3}{32}, \qquad \text{fail: } \tfrac{1}{3} \cdot \tfrac{1}{4} \cdot \tfrac{1}{4} \cdot \tfrac{3}{4} = \tfrac{1}{64}.$$

Normalizing, $\widehat{\Pr}(\text{pass} \mid x) = \frac{3/32}{3/32+1/64} = \frac{6}{7} = 0.857$.

(c) Pass: $\frac{2}{3} \cdot \frac{2}{8} \cdot \frac{1}{8} \cdot \frac{3}{8} = \frac{1}{128}$. Fail: $\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{64} = \frac{18}{128}$. So $\widehat{\Pr}(\text{pass} \mid x) = 1/19 = 0.053$.

(d) No failing student did all the problems, so $\hat{\Pr}(\text{all} \mid \text{fail}) = 0$ and the fail product is 0 whatever else C did: $\hat{\Pr}(\text{pass} \mid x) = 1$. A single zero count overrides every other predictor, and C's other two values both point toward fail. C's combination (no, all, no) is not in the training data, but naive Bayes doesn't need it to be: it uses each predictor separately. The problem is one empty cell in one table.

(e) Smoothed, for pass ($n_k = 8$): Pr(attended no) = 3/10, Pr(all) = 4/11, Pr(slept no) = 4/10. For fail ($n_k = 4$): 4/6, 1/7 and 4/6. Then

$$\text{pass: } \tfrac{2}{3} \cdot \tfrac{3}{10} \cdot \tfrac{4}{11} \cdot \tfrac{4}{10} = 0.0291, \qquad \text{fail: } \tfrac{1}{3} \cdot \tfrac{4}{6} \cdot \tfrac{1}{7} \cdot \tfrac{4}{6} = 0.0212,$$

and $\hat{\Pr}(\text{pass} \mid x) = 0.0291/(0.0291 + 0.0212) = 0.579$. Doing all the problems still pushes toward pass, but no longer decides it alone. (This matches sklearn's default, `CategoricalNB(alpha=1)`.)

(f) The predictors have $2 \times 3 \times 2 = 12$ combinations, so the joint distribution needs $12 - 1 = 11$ probabilities per class. Naive Bayes needs $1 + 2 + 1 = 4$ per class (one for each binary predictor, two for the three-level one). With only 4 failing students, most of the 12 cells in the joint table would be empty, so most of its estimates would be 0 or 0/0. With 30 binary predictors the full table would need about 10^9 probabilities per class; naive Bayes needs 30.

5. **What naive Bayes assumes, and when it works.** (ISL 4.4.4, 4.5.) Answer each in a few sentences.

(a) State the naive Bayes assumption in words. Is it about how the predictors relate to each other in the population as a whole, or within each class? Give an example of two predictors that are strongly correlated in the population but roughly independent within each class.

(b) The naive Bayes assumption is rarely true, yet naive Bayes often classifies well. Explain why, in terms of bias and variance. In what kind of problem does it tend to beat QDA?

(c) A spam filter uses the words "free" and "offer", which almost always appear together in spam. How does naive Bayes treat the two words, and what does that do to its estimate of Pr(spam $\mid x$) for an email containing both? Does it necessarily change which class is predicted?

(d) True or false: Gaussian naive Bayes, with a separate variance for each predictor in each class, has a linear decision boundary.

Solution.

(a) Within each class, the predictors are independent, so the class density is a product of one-dimensional densities: $f_k(x) = \prod_{j=1}^{p} f_{kj}(x_j)$. It is about independence *within each class*, not overall. Example: let the classes be children and adults, with X_1 = height and X_2 = vocabulary size. Across everyone the two are strongly correlated, because adults have more of both, but among adults alone (or children of one age) height says little about vocabulary. Naive Bayes is fine there.

(b) The assumption adds bias: when it is false, naive Bayes's estimates of $p_k(x)$ are wrong. But classification only needs the right class to have the largest estimate, and biased

probabilities are often still on the right side of the boundary. In return, naive Bayes estimates only p one-dimensional densities per class (for Gaussian naive Bayes, $2p$ numbers per class against QDA's $p + p(p+1)/2$), so its variance is much lower. The reduction in variance tends to outweigh the bias when n is small relative to p, which is where naive Bayes beats QDA. With large n and correlated predictors, QDA's lower bias wins.

(c) It treats them as two independent pieces of evidence and multiplies in both likelihood ratios, $\hat{f}_{\text{spam}}(\text{free})/\hat{f}_{\text{ham}}(\text{free})$ and the same for "offer". Since the words carry largely the same information, the evidence is counted about twice, and $\widehat{\Pr}(\text{spam} \mid x)$ is pushed too close to 1: the filter is overconfident. It doesn't necessarily change the prediction, since both words point toward spam anyway. But for a borderline email the double-counted evidence can outweigh the prior or the other words and flip it, and any rule that uses the size of the probability (say, delete only when it is above 0.99) is affected directly.

(d) False. Each predictor contributes $-(x_j - \mu_{kj})^2/2\sigma_{kj}^2 - \log \sigma_{kj}$ to $\delta_k(x)$, so as in Problem 1 the boundary between classes k and l has an x_j^2 term with coefficient $\frac{1}{2}(1/\sigma_{lj}^2 - 1/\sigma_{kj}^2)$: it is quadratic. Unlike QDA's boundary, it has no cross terms $x_j x_m$, because naive Bayes models each predictor separately. It is linear only if every predictor has the same variance in both classes, and then it is LDA with a diagonal covariance matrix.